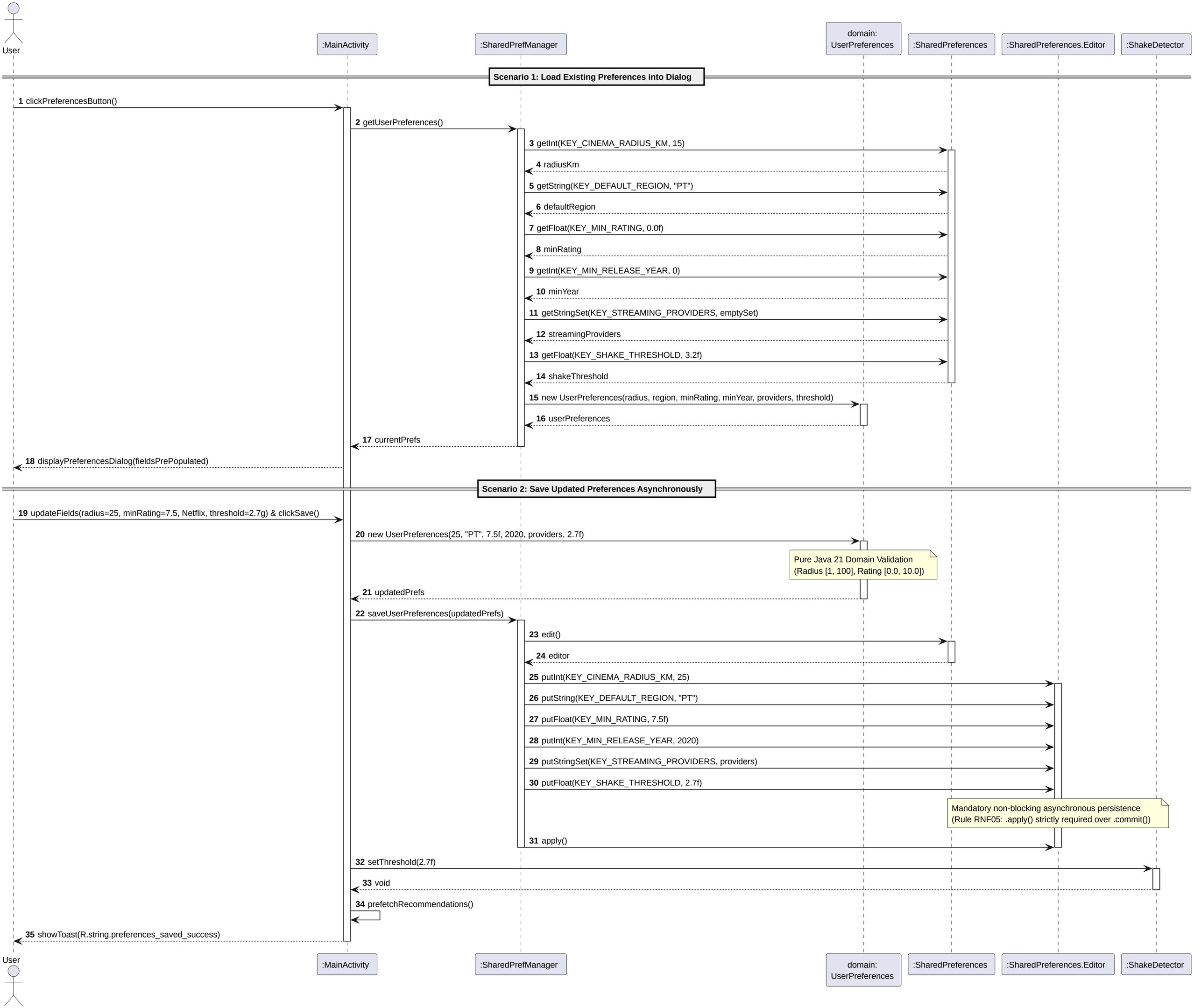
User
:MainActivity
:SharedPrefManager
domain:
UserPreferences
:SharedPreferences
:SharedPreferences.Editor
:ShakeDetector
Scenario 1: Load Existing Preferences into Dialog
1 clickPreferencesButton()
2 getUserPreferences()
3 getInt(KEY_CINEMA_RADIUS_KM, 15)
4 radiusKm
5 getString(KEY_DEFAULT_REGION, "PT")
6 defaultRegion
7 getFloat(KEY_MIN_RATING, 0.0f)
8 minRating
9 getInt(KEY_MIN_RELEASE_YEAR, 0)
10 minYear
11 getStringSet(KEY_STREAMING_PROVIDERS, emptySet)
12 streamingProviders
13 getFloat(KEY_SHAKE_THRESHOLD, 3.2f)
14 shakeThreshold
15 new UserPreferences(radius, region, minRating, minYear, providers, threshold)
16 userPreferences
17 currentPrefs
18 displayPreferencesDialog(fieldsPrePopulated)
Scenario 2: Save Updated Preferences Asynchronously
19 updateFields(radius=25, minRating=7.5, Netflix, threshold=2.7g) & clickSave()
20 new UserPreferences(25, "PT", 7.5f, 2020, providers, 2.7f)
Pure Java 21 Domain Validation
(Radius [1, 100], Rating [0.0, 10.0])
21 updatedPrefs
22 saveUserPreferences(updatedPrefs)
23 edit()
24 editor
25 putInt(KEY_CINEMA_RADIUS_KM, 25)
26 putString(KEY_DEFAULT_REGION, "PT")
27 putFloat(KEY_MIN_RATING, 7.5f)
28 putInt(KEY_MIN_RELEASE_YEAR, 2020)
29 putStringSet(KEY_STREAMING_PROVIDERS, providers)
30 putFloat(KEY_SHAKE_THRESHOLD, 2.7f)
Mandatory non-blocking asynchronous persistence
(Rule RNF05: .apply() strictly required over .commit())
31 apply()
32 setThreshold(2.7f)
33 void
34 prefetchRecommendations()
35 showToast(R.string.preferences_saved_success)
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